

Name: _____





BOOTSTRAP

Equity • Scale • Rigor

Brought to you by the Bootstrap team:

- Emmanuel Schanzer
- Kathi Fiser
- Shriram Krishnamurthi
- Dorai Sitaram
- Joe Politz
- Ben Lerner
- Nancy Pfenning
- Flannery Denny
- Rachel Tabak

Supervised Learning

For this page, you'll need [Image Similarity Starter File](#) open on your computer

How good is our **angle-similarity** measure at finding similar images for snow, grass, and sun?

1) Evaluate each of the expressions below, paying attention to the closest matches at the top of the result table

```
(a) Snow: angle-similarity(computed, "snow2", [list: "LUMINANCE", "ENTROPY", "red", "green", "blue"])
```

(b) **Grass:** angle-similarity(computed, "grass", [list: "LUMINANCE", "ENTROPY", "red", "green", "blue"])

```
(c) Sun: angle-similarity(computed, "sun2", [list: "LUMINANCE", "ENTROPY", "red", "green", "blue"])
```

Were the top results right for images with snow? _____ Grass? _____ Sun? _____

Why is our similarity method worse at recommending sun images than it is with snow or grass? _____

Calculating Centroids

2) Table below includes two rows with images of the sun ("grass-sun" and "sun2"). Complete the table by filling the Centroid.

(NOTE: we're just showing *some* of the columns on this page)

ID	WIDTH	HEIGHT	red	green	blue
"grass-sun"	160	107	8932	5245	2921
"sun2"	160	107	10971	175	5908
"SUN CENTROID"					

3) Evaluate `m-sun = add-centroid(computed, "SUN", [list: "grass-sun", "sun2"])` to define a new model with the

centroid, then `m-sun` to see it. What do you get back? (HINT: Be sure to view all the rows `m-sun` !)

4) Complete the code below, to find similar images using the "SUN CENTROID", then try it out in Pyret:

```
angle-similarity(table, ID to compare, [list: "LUMINANCE", "ENTROPY", "red", "green", "blue"]).
```

Searching and Recommending

The questions below rely on the following contracts:

- `# search-by-tag :: ( Tablet, Stringtag ) -> Table`
- `# recommend :: ( Tablet ) -> Table`

5) Make sure that only "grass-sun" and "sun2" are tagged with "sun", click "Run", and evaluate

`search-by-tag( computed, "sun" )`. What does `search-by-tag` do?

6) Try tagging **one** of the grass images. Does `search-by-tag(computed, "grass")` recommend the other? _____ What about for snowy images? _____

7) Try adding a small number of likes and dislikes to the table, then evaluate `recommend( computed )`. The `recommend` function builds two centroids. Can you figure out why?

Supervised or Unsupervised?

Machine learning models are all around us, using a mix of different learning methods. Can you tell the difference between **supervised learning** and **unsupervised learning**? Supervised methods learn from labeled examples — sorting things into categories (*classification*), predicting a number (*regression*), or recommending by similarity to examples you've tagged (building a *centroid*). Unsupervised methods hunt for patterns with no labels at all.

Example	What Kind of Learning and Why? (Circle one)																						
1) An e-commerce website uses historical data of past transactions, which are explicitly marked as 'authorized' or 'fraudulent', to train a model that flags suspicious new transactions.	<table border="1"> <thead> <tr> <th colspan="3">Supervised</th> <th>Unsupervised</th> </tr> <tr> <td>Classification</td> <td>Regression</td> <td>Similarity</td> <td></td> </tr> </thead> <tbody> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> </tbody> </table>			Supervised			Unsupervised	Classification	Regression	Similarity													
Supervised			Unsupervised																				
Classification	Regression	Similarity																					
2) A hospital uses a machine learning model to classify medical images (X-rays) as either 'healthy' or 'pneumonia-affected' after training it on a dataset of images reviewed and tagged by expert radiologists.	<table border="1"> <thead> <tr> <th colspan="3">Supervised</th> <th>Unsupervised</th> </tr> <tr> <td>Classification</td> <td>Regression</td> <td>Similarity</td> <td></td> </tr> </thead> <tbody> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> </tbody> </table>			Supervised			Unsupervised	Classification	Regression	Similarity													
Supervised			Unsupervised																				
Classification	Regression	Similarity																					
3) A real estate app estimates the market price of a house based on features like square footage, number of bedrooms, and zip code, using a model trained on thousands of past home sales with known prices.	<table border="1"> <thead> <tr> <th colspan="3">Supervised</th> <th>Unsupervised</th> </tr> <tr> <td>Classification</td> <td>Regression</td> <td>Similarity</td> <td></td> </tr> </thead> <tbody> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> </tbody> </table>			Supervised			Unsupervised	Classification	Regression	Similarity													
Supervised			Unsupervised																				
Classification	Regression	Similarity																					
4) A music app lets you "heart" the songs you love. It averages the audio features (tempo, energy, danceability) of all your hearted songs into a single "taste profile," then recommends new songs whose features land closest to that profile.	<table border="1"> <thead> <tr> <th colspan="3">Supervised</th> <th>Unsupervised</th> </tr> <tr> <td>Classification</td> <td>Regression</td> <td>Similarity</td> <td></td> </tr> </thead> <tbody> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> </tbody> </table>			Supervised			Unsupervised	Classification	Regression	Similarity													
Supervised			Unsupervised																				
Classification	Regression	Similarity																					
5) A marketing team splits their customer base into groups so they can design targeted ad campaigns, based only on data they've collected: like purchasing frequency, average spend, and age.	<table border="1"> <thead> <tr> <th colspan="3">Supervised</th> <th>Unsupervised</th> </tr> <tr> <td>Classification</td> <td>Regression</td> <td>Similarity</td> <td></td> </tr> </thead> <tbody> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> </tbody> </table>			Supervised			Unsupervised	Classification	Regression	Similarity													
Supervised			Unsupervised																				
Classification	Regression	Similarity																					
6) A streaming platform recommends new movies to a user based only on what they've seen before (not what they've liked or disliked), and does not use any labeled information about the genre, cast, etc.	<table border="1"> <thead> <tr> <th colspan="3">Supervised</th> <th>Unsupervised</th> </tr> <tr> <td>Classification</td> <td>Regression</td> <td>Similarity</td> <td></td> </tr> </thead> <tbody> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> <tr><td colspan="4"> </td></tr> </tbody> </table>			Supervised			Unsupervised	Classification	Regression	Similarity													
Supervised			Unsupervised																				
Classification	Regression	Similarity																					

Comparing Desserts

Dessert Attributes

1) Add *two examples* to the table, filling in values for each attribute.

ID	Dessert	crunchiness 0 = not, 3 = very	chocolatiness 0 = not, 3 = totally	ideal temperature 1 = frozen, 5 = hot	
1	s'mores	2	1	4	
2	walnut chocolate chip cookie	1	1	4	
3	berry froyo	0	0	1	
4					
5					

2) Think of one more attribute you could use to compare the 5 desserts on the list and fill in the last column.

3) Put a star (★) next to your favorite dessert on the list.

4) Put a triangle (△) next to the other example that is the *most similar* to your favorite dessert.

5) How could you **measure the similarity** between any two examples? _____

Graph Your Space

6) **Label** and **number** your axes on the two graphs below, paying attention to the scale of each attribute. On the second graph, compare an attribute of your choosing to Chocolatiness.



7) Plot your examples as points in both graphs, using a star (★) to plot your favorite example and a triangle (△) for the example you selected as most similar above.

8) Is your choice for "most similar" (△) the closest to your favorite (★) in the **first graph**? _____ In the **second graph**? _____

9) Do you have any **new** ideas for how to measure the similarity between any two desserts? _____

These materials were developed partly through support of the National Science Foundation (awards 1042210, 1535276, 1648684, and 1738598) and are licensed under a Creative Commons 4.0 Unported License. Based on a work at www.BootstrapWorld.org. Permissions beyond the scope of this license may be available by contacting contact@BootstrapWorld.org.